

Nilpotents of Every Index for White's Lacunary Weight

Run `fa_banach_001`, lane 0 (`agent_lane_00`)

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Abstract

White constructs the weight $\omega(n) = \exp(|n|_{S_0})$ on $\mathbb{Z}$, where $S_0 = \{2^{i^2} : i \geq 0\}$, and asks whether the radical of $\ell^1(\mathbb{Z}, \omega)^{**}$ contains nilpotent elements of arbitrarily high index. We answer yes. In fact, for every $m \geq 2$ we construct a weak-star cluster point Φ_m of normalized point masses such that Φ_m has nilpotency index exactly m . More strongly, the principal left ideal $\ell^1(\mathbb{Z}, \omega)^{**} \square \Phi_m$ has nilpotency index exactly m . The construction uses sparse balanced digits in the mixed-radix chain $2^{i^2} \mid 2^{(i+1)^2}$: fewer than m copies create no carries, whereas m copies create an independent word-length saving at every common digit.

Status: candidate full result, likely valid. The literal open question following Theorem 1.3 of arXiv:1708.09635 is resolved affirmatively for the paper's exact weight. Novelty is plausible but not certified; human review should focus on the balanced-digit lemma and the passage from the uniform convolution estimate to the Arens product.

1 Exact source question

Let

$$q_i = 2^{i^2} \quad (i \geq 0), \quad S_0 = \{q_i : i \geq 0\}, \quad \eta(n) = |n|_{S_0}, \quad \omega(n) = e^{\eta(n)}.$$

Here $\eta(n)$ is the least number of signed elements of S_0 whose sum is n . White proves that the radical of A^{**} , where $A = \ell^1(\mathbb{Z}, \omega)$ and A^{**} carries the first Arens product $\square$, contains a nonnilpotent element. He then asks [1, p. 2]:

*rad($L^1(G)^{**}$) = {0}, for G a locally compact group. It also seems that until now it was not known whether or not $\text{rad}(L^1(G, \omega)^{**})$ is always nilpotent, for G a locally compact group and ω a weight on G , although there is an example of a weight on $\mathbb{Z}$ in [5, Example 9.15] for which this radical cubes to zero, but has non-zero square. In Section 4 we shall answer both of these questions in the negative by proving the following:*

Theorem 1.2. *Let $G = \oplus_{i=1}^{\infty} \mathbb{Z}$. Then $\text{rad}(\ell^1(G)^{**})$ contains nilpotent elements of every index.*

Here we understand $\oplus_{i=1}^{\infty} \mathbb{Z}$ to consist of integer sequences which are eventually zero, so that our example is a countable abelian group.

We note that by a theorem of Grabiner [9], Theorem 1.2 implies that $\text{rad}(\ell^1(\oplus_{i=1}^{\infty} \mathbb{Z})^{**})$ contains non-nilpotent elements. In Section 5, we obtain a similar result on $\mathbb{Z}$, but this time involving a weight.

Theorem 1.3. *There exists a weight ω on $\mathbb{Z}$ such that $\text{rad}(\ell^1(\mathbb{Z}, \omega)^{**})$ contains non-nilpotent elements.*

However, we do not know whether or not this example has nilpotent elements of arbitrarily high index.

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We prove the stronger statement that the radical contains a principal left ideal of every prescribed finite nilpotency index.

2 Idea of the proof

Proof Intuition

The generators q_i form a mixed-radix system with enormous even radices

$$R_i = \frac{q_{i+1}}{q_i} = 2^{2^{i+1}}.$$

Fix the desired index m . At sparse positions r_j , put a digit d_j just larger than $R_{r_j}/(2m)$, and let t_k be the integer formed from the first k such digits. If fewer than m prefixes are added, every resulting digit remains strictly below half its radix, so the word lengths add exactly. With m prefixes, every position common to all of them crosses the half-radix threshold. Replacing md_j copies of q_{r_j} by one q_{r_j+1} and a small negative digit saves at least one generator. Sparsity keeps these carries disjoint. Hence the normalized product of m point masses has norm at most $e^{-\min(k_1, \dots, k_m)}$, while every product of fewer than m has norm one.

A weak-star cluster point Φ_m therefore satisfies $\Phi_m^m = 0$ but $\Phi_m^{m-1} \neq 0$. The same estimate survives after arbitrary bounded factors are inserted between the point masses: at the finite level convolution on $\mathbb{Z}$ is commutative, so those factors can be grouped separately. Consequently the entire left ideal $A^{**} \square \Phi_m$ vanishes at the chosen power m , placing Φ_m in the Jacobson radical.

3 A balanced mixed-radix lemma

We use the following elementary exact description of the word metric.

Lemma 1 (strict balanced digits). *Suppose a finite expansion*

$$x = \sum_{i \geq 0} c_i q_i$$

satisfies

$$|c_i| \leq \frac{R_i}{2} - 1 \quad (i \geq 0).$$

Then

$$\eta(x) = \sum_{i \geq 0} |c_i|.$$

Proof. The displayed expansion gives the upper bound. Consider any other finite integer-coefficient representation $x = \sum_i a_i q_i$. Normalize it from low positions to high positions. Once positions below i agree with the digits $c_0, \dots, c_{i-1}$, reduction modulo $q_{i+1} = R_i q_i$ gives $a_i = c_i + R_i h$ for some $h \in \mathbb{Z}$. Replace

$$(a_i, a_{i+1}) \quad \text{by} \quad (c_i, a_{i+1} + h).$$

This preserves the represented integer. It does not increase the sum of absolute coefficients, because

$$|c_i| + |h| \leq |c_i + R_i h|.$$

Indeed, this is immediate when $c_i h \geq 0$. When $c_i h < 0$, we have

$$|c_i + R_i h| - |c_i| - |h| = (R_i - 1)|h| - 2|c_i| \geq (R_i - 1) - (R_i - 2) > 0.$$

Iterating through the finitely many nonzero positions converts the arbitrary representation into the displayed one without increasing its length. Thus the displayed length is minimal. $\square$

4 Full resolution

Theorem 1. *For the exact weight $\omega(n) = \exp(|n|_{S_0})$ constructed in [1], and for every integer $m \geq 2$, there exists*

$$\Phi_m \in \text{rad}(\ell^1(\mathbb{Z}, \omega)^{**}, \square)$$

whose nilpotency index is exactly m . More strongly, the principal left ideal

$$I_m = \ell^1(\mathbb{Z}, \omega)^{**} \square \Phi_m$$

has nilpotency index exactly m .

Proof. Fix $m \geq 2$. Choose $N \geq 2$ so large that $R_i \geq 4m^2$ for every $i \geq N$, and set

$$r_j = N + 2(j - 1), \quad d_j = \left\lfloor \frac{R_{r_j}}{2m} \right\rfloor + 1 \quad (j \geq 1).$$

The choice of N gives

$$(m - 1)d_j \leq \frac{R_{r_j}}{2} - 1, \quad md_j \geq \frac{R_{r_j}}{2} + 1. \quad (1)$$

Indeed, the second inequality is immediate, while

$$(m - 1)d_j \leq \frac{(m - 1)R_{r_j}}{2m} + m - 1 = \frac{R_{r_j}}{2} - \frac{R_{r_j}}{2m} + m - 1 \leq \frac{R_{r_j}}{2} - 1.$$

Define

$$t_k = \sum_{j=1}^k d_j q_{r_j} \quad (k \geq 1).$$

Lemma 1 and (1) show that

$$\eta(t_k) = \sum_{j=1}^k d_j. \quad (2)$$

Take $1 \leq s \leq m$ and positive integers $k_1, \dots, k_s$. In the sum $t_{k_1} + \dots + t_{k_s}$, the coefficient at q_{r_j} is $h_j d_j$, where

$$h_j = |\{\ell : j \leq k_\ell\}|.$$

If $s < m$, then $h_j \leq m - 1$, so all digits are strictly balanced by (1). Lemma 1 therefore gives

$$\eta(t_{k_1} + \dots + t_{k_s}) = \sum_{\ell=1}^s \eta(t_{k_\ell}) \quad (s < m). \quad (3)$$

Now let $s = m$, and write $K = \min(k_1, \dots, k_m)$. For every $j \leq K$, replace the digit md_j by

$$md_j q_{r_j} = (md_j - R_{r_j}) q_{r_j} + q_{r_{j+1}}.$$

The new digit at r_j has absolute value at most $R_{r_j}/2 - 1$, by the second inequality in (1); the digit at $r_j + 1$ is 1. The positions r_j are separated by two, so no carry position is another selected position. All remaining digits have multiplicity at most $m - 1$. Thus the resulting expansion is strictly balanced, and Lemma 1 gives the exact identity

$$\eta(t_{k_1} + \cdots + t_{k_m}) = \sum_{\ell=1}^m \eta(t_{k_\ell}) - \sum_{j=1}^K \sigma_j, \quad \sigma_j = 2md_j - R_{r_j} - 1 \geq 1. \quad (4)$$

Put $e_t = \delta_t/\omega(t)$. Then $\|e_t\|_\omega = 1$, and

$$\|e_{u_1} * \cdots * e_{u_s}\|_\omega = \exp\left(\eta(u_1 + \cdots + u_s) - \sum_{\ell=1}^s \eta(u_\ell)\right). \quad (5)$$

Equations (3)–(4) imply

$$\|e_{t_{k_1}} * \cdots * e_{t_{k_s}}\|_\omega = 1 \quad (s < m), \quad (6)$$

$$\|e_{t_{k_1}} * \cdots * e_{t_{k_m}}\|_\omega \leq e^{-K} \quad (s = m). \quad (7)$$

Choose a weak-star convergent subnet of (e_{t_k}) in the unit ball of A^{**} , and call its limit Φ_m . Since the weight ω itself is a norm-one functional in $A^* = \ell^\infty(\mathbb{Z}, 1/\omega)$, equation (6) and the iterated-limit definition of $\square$ give

$$\langle \Phi_m^{\square s}, \omega \rangle = 1 \quad (1 \leq s < m).$$

In particular $\Phi_m^{\square(m-1)} \neq 0$. On the other hand, (7) tends to zero uniformly whenever all k_ℓ tend to infinity. Every iterated weak-star limit defining $\Phi_m^{\square m}$ is therefore zero, so

$$\Phi_m^{\square m} = 0.$$

Thus Φ_m has nilpotency index exactly m .

It remains to prove radical membership. Let $\Psi_1, \dots, \Psi_m \in A^{**}$, and choose bounded nets $(a_{i,\alpha}) \subset A$ converging weak-star to Ψ_i . Expand

$$(\Psi_1 \square \Phi_m) \square \cdots \square (\Psi_m \square \Phi_m)$$

using the iterated-limit definition of the first Arens product. At every finite stage the approximating product is, because convolution on $\mathbb{Z}$ is commutative,

$$(a_{1,\alpha_1} * \cdots * a_{m,\alpha_m}) * (e_{t_{k_1}} * \cdots * e_{t_{k_m}}).$$

Its norm is at most

$$\left(\prod_{i=1}^m \|a_{i,\alpha_i}\|_\omega\right) e^{-K}$$

by (7). The nets $(a_{i,\alpha})$ are bounded and the estimate is uniform in their indices, so every required iterated limit is zero. Hence $I_m^{\square m} = \{0\}$. Conversely, $\Phi_m \in I_m$ because A^{**} is unital, and $\Phi_m^{\square(m-1)} \neq 0$, so I_m has index exactly m .

A nilpotent left ideal consists of quasiniptotent elements and is contained in the Jacobson radical. Therefore $\Phi_m \in \text{rad}(A^{**}, \square)$, as claimed. $\square$

5 Verification and boundary checks

The proof uses strict inequalities, so there is no ambiguous half-radix digit. The spacing $r_{j+1} - r_j = 2$ is essential: it ensures that the carry q_{r_j+1} does not alter another chosen digit. The construction also uses prefixes rather than disjoint blocks; consequently exactly the first $\min(k_1, \dots, k_m)$ digits carry, producing the uniform decay needed for every ordering of Arens limits.

The accompanying script independently minimizes finite word-length problems by mixed-integer linear programming. It checks all threshold formulas for $m = 2, \dots, 8$, 32 exact word-length identities, and 69,632 instances of the local normalization inequality. These checks are corroborative only; Lemma 1 supplies the proof for all m .

6 Novelty audit and limitations

The local run indexes, exact-title and exact-question web searches, phrase variants involving weighted $\ell^1(\mathbb{Z})$, the source author's current publication list, and the author's 2026 sequel arXiv:2602.02764 [2] were checked through 17 August 2026. The sequel concerns different radicals for the two Arens products on a weighted free group and does not state this result. The OpenAlex record for the 2018 source reported no indexed citing works at the time of the audit. No prior answer to this exact question was located. This is a bounded search, not a priority claim.

The theorem concerns the particular lacunary weight in the question and the first Arens product, exactly as asked. It does not settle the source's much broader open questions about $\text{rad}(\ell^1(\mathbb{Z})^{**})^2$, semisimplicity for nonamenable groups, or Fourier algebras.

References

- [1] J. T. White, *The radical of the bidual of a Beurling algebra*, Quarterly Journal of Mathematics **69** (2018), 975–993; arXiv:1708.09635.
- [2] J. T. White, *Radicals of Biduals of Beurling Algebras Can Be Different for the Two Arens Products*, arXiv:2602.02764 (2026).